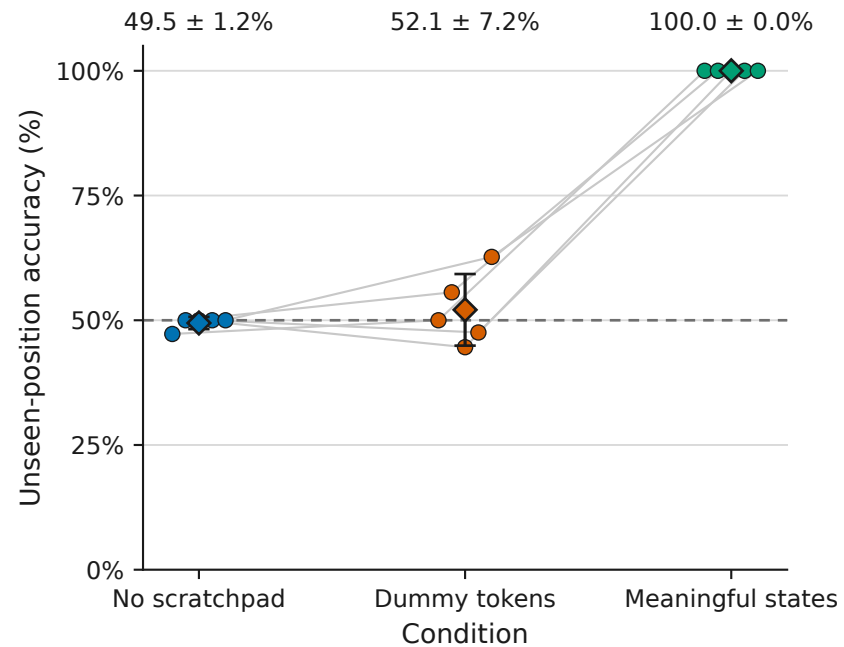


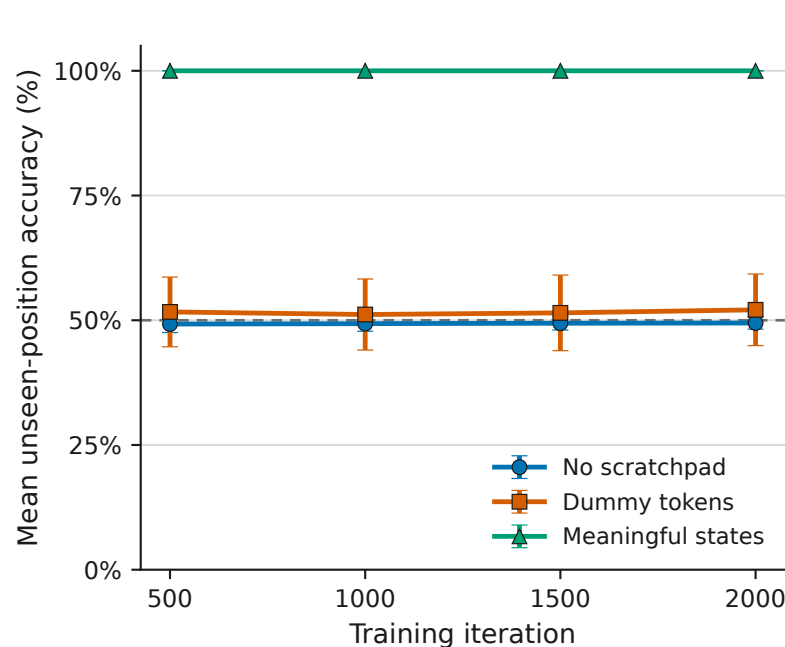
Formal Scratchpad Generalization Result

Five paired model seeds; RoPE; fixed input length = 20; free-running greedy inference

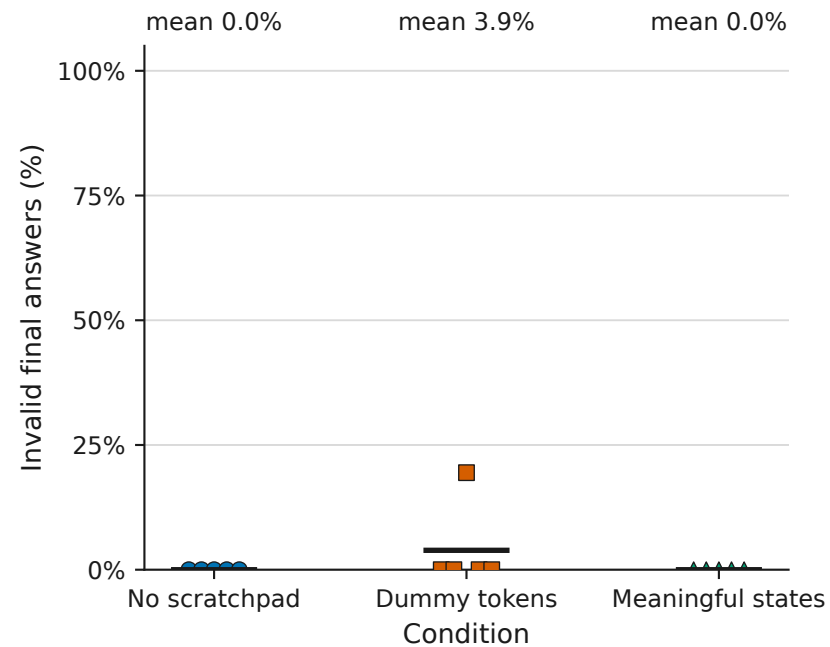
A Final accuracy (2,000 iterations)



B Accuracy across checkpoints



C Invalid outputs at 2,000 iterations



Points: individual seeds; diamond/bar: mean; error bars: sample SD; dashed line: 50% chance

Seen-position accuracy at 2,000 iterations: 100% for every condition and seed